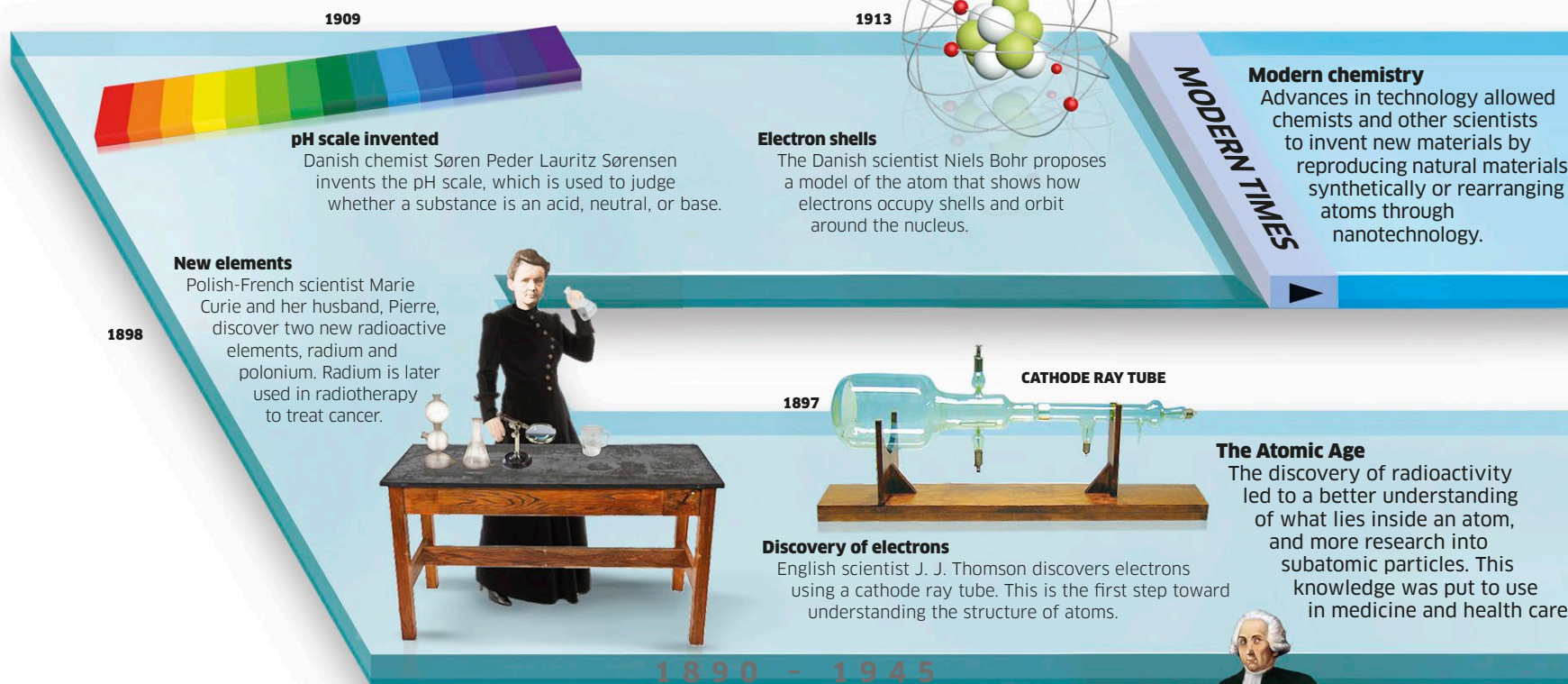




Discovering matter

Thousands of years of questioning, experimentation, and research have led to our understanding of matter as we know it today.

Following the earliest explorations of matter by our prehistoric ancestors, Greek philosophers were among the first people to attempt to classify matter and explain its behavior. Over time, scientists found more sophisticated ways of analyzing different types of matter and discovered many of the elements. The Industrial Revolution saw the invention of new synthetic materials using these elements, while greater understanding of the structure of atoms led to significant advances in medicine. New substances and materials with particularly useful properties are still being discovered and invented to this day.

Timeline of discoveries

From prehistory to the present day, people have sought to understand how matter behaves and to classify different types. Over the years, this has led to the discovery of new matter and materials.

Prehistory to antiquity

The earliest discoveries of how matter behaves were made not by scientists, but by prehistoric ancestors trying to survive. During antiquity, philosophers spent a lot of time trying to work out what matter is.



Making fire

Our ancestors learn to make fire using combustion (although they don't know that at the time).

Copper and bronze

Smelting of copper (extracting it from its ore through heat) is discovered. Bronze (copper smelted with tin) is first produced in 3200 BCE.



Greek philosophers

Empedocles suggests that everything is made of four elements: air, earth, fire, and water. Democritus suggests that all matter consists of atoms.



1772 / 1774

Discovery of oxygen

Swedish chemist Carl Scheele builds a contraption to capture oxygen by heating various compounds together. English scientist Joseph Priestley also discovers oxygen by showing that a candle can't burn without it.



SHEELER'S OXYGEN APPARATUS

1789

Antoine Lavoisier

French chemist Antoine Lavoisier publishes *Elements of Chemistry*, which lists the 33 known elements divided into four types: gases, metals, non-metals, and earths.



BEFORE 500 CE

790,000 BCE

3200 BCE

420 BCE